Lecture 2

Tuesday, 17 January 2023 12:00 PM

$c_j = \frac{\rho_j}{m_j}$; $J_j = \frac{j_j}{m_j}$ = molar flux relative to v

perfect mixing → Don't need momentum balance

Imperfect mixing → Residence time distribution

Continuity eq:

$$\frac{\partial c_j}{\partial t} + v \cdot \nabla c_j + \nabla \cdot J_j + c_j(\nabla \cdot v) = \sum_i \alpha_{ij} r_i$$

→ Rowing across the river (fishing) ⟶ gives total derivative (material derivative)

→ Fishing at the spot → Local derivative

Heterogenous reactions

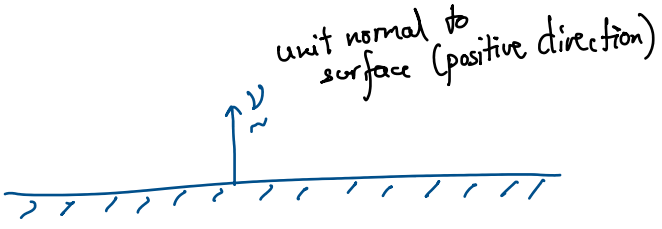
$$\frac{Dc_j}{Dt} + \nabla \cdot J_j = \sum_j^m \alpha_{ij} a_v \hat{r}_i - c_j \frac{D \ln \hat{V}}{Dt}$$

Transient term; Diffusive term

a_v = active per unit volume

$\hat{r}$ = reaction per unit area of catalyst surface.

Reaction located at a specific part ⟶ occurs at a boundary



Intrinsic reaction rate is always positive

$\alpha_j > 0$ (for reactant)

↳ going to a catalyst surface

rate at which the reaction is happening

positive

↳ going to a catalyst surface

rate at which reaction is happening

Consuming solid surface

$\partial\Omega_t \rightarrow \partial\Omega_{t+dt}$

↑ shape of the surface at time t

$x - \frac{\nabla f(x,t)}{\|\nabla f(x,t)\|} \hat{r}\, dt$

Normal to a surface

Represents a location on the solid surface

Leading to change in surface shape

Chemical kinetics:

→ Mass action kinetics ⇒ (of elementary reactions)

$A \rightarrow B$

$r = k[C_A]^1$

→ Homogenous reaction: + deDonder kinetics

$\sum_{j=1}^{s} \alpha_j A_j = 0$ ← given reaction sequences.

Batch: $c_j = c_{j_0} + \alpha_j \xi$ (No change in volume)

How the reaction rate changes as ξ changes

$$r(c_j, T) = k \prod_{j=1}^{s} (c_{j_0} + \alpha_j \xi)^{a_j} - k' \prod_{j=1}^{s} (c_{j_0} + \alpha_j \xi)^{a'_j}$$

reverse reaction

$$\underset{\sim}{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_j \end{bmatrix}$$

$$\left.\frac{\partial r}{\partial \xi}\right|_T = ?$$

We define:

$a_j \equiv \frac{1}{2}(|\alpha_j| - \alpha_j)$; $\alpha_j < 0$ (reactant) ⇒ $a_j = -\alpha_j$

$|\alpha_j| = -\alpha_j$

or

$a_j = 0$ (if $\alpha_j \geq 0$)

$a_j = 0$ α_j [illegible]

Similarly:

$$a_j' = \frac{1}{2}\left(|\alpha_j| + \alpha_j\right) = \begin{cases} 0 & \alpha_j < 0 \\ \alpha_j & \alpha_j > 0 \end{cases}$$

Now we differentiate:

$$\left.\frac{\partial r}{\partial \xi}\right|_T = \sum_{r=1}^{S} \frac{\partial c_r}{\partial \xi} \frac{\partial r}{\partial c_r}$$

$$= \sum_{r=1}^{S} k a_r \left(\prod_{j=1}^{S} c_j^{a_j} \right)$$